

Laplace Distribution (Double exponential Distribution):

A continuous r.v. X is said to be followed a Laplace distribution if its pdf is given as;

$$f(x) = \frac{1}{2\theta} e^{-\frac{|x-\mu|}{\theta}}, \quad -\infty < x < \infty, \quad -\infty < \mu < \infty, \quad \theta > 0$$

The continuous random variable representing long life time of machines or objects follows Laplace distribution.

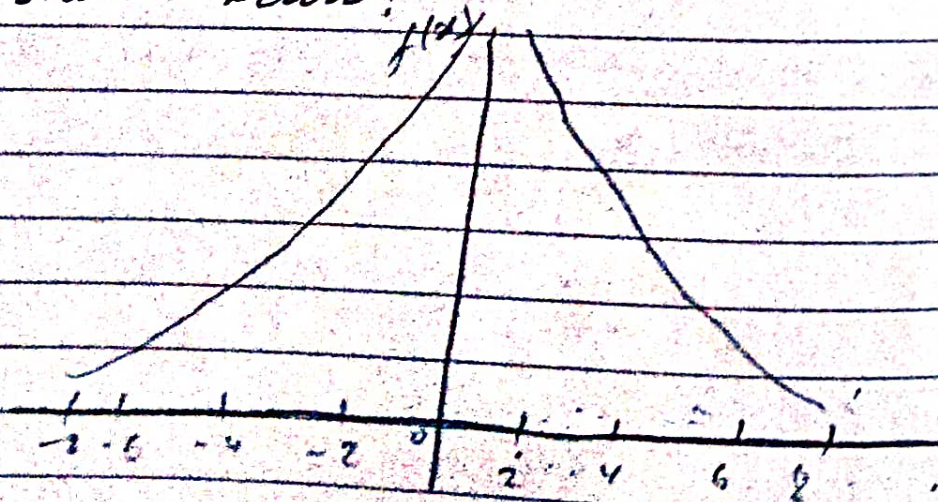
If $\mu=0$ and $\theta=1$ then the r.v. X has a standard Laplace distribution defined as follows:

A random variable X is said to be followed a standard Laplace distribution if its pdf is defined as;

$$f(x) = \frac{1}{2} e^{-|x|}, \quad -\infty < x < \infty$$

$$\therefore X \sim L(0, 1).$$

The graph of the standard Laplace pdf is shown below;



Moments of Standard Laplace Distribution:

Let $X \sim L(0,1)$ then its r th moment about origin of X is given as

$$\mu_r' = E(X^r) = \int_{-\infty}^{\infty} x^r f(x) dx$$

$$= \frac{1}{2} \int_{-\infty}^{\infty} x^r e^{-|x|} dx$$

$$= \frac{1}{2} \left[\int_{-\infty}^0 x^r e^x dx + \int_0^{\infty} x^r e^{-x} dx \right]$$

Here, $\int_{-\infty}^0 x^r e^x dx$

put $x = -y$

$dx = -dy$

$$= \int_{-\infty}^0 (-y)^r e^{-y} (-dy)$$

$$= (-1)^r \left[\int_0^{\infty} y^r e^{-y} dy \right]$$

$$= (-1)^r \Gamma(r+1)$$

$$= (-1)^r r!$$

$$\therefore \mu_r' = \frac{1}{2} \left[(-1)^r r! + r! \right]$$

$$\boxed{\mu_r' = \frac{[(-1)^r + 1] r!}{2}}$$

Put $r = 1, 2, 3, 4,$

$$\mu_1' = 0, \quad \mu_2' = 2, \quad \mu_3' = 0 \quad \text{and} \quad \mu_4' = 24$$

Since, $\mu_1' = 0$, the central moments of standard Laplace distribution coincide with the raw moment.

$$\mu_1 = 0, \mu_2 = 2, \mu_3 = 0, \mu_4 = 24$$

Hence, the standard Laplace distribution has mean = 0, and variance = 2

$$\text{Skewness } (\beta_1) = \frac{\mu_3^2}{\mu_2^3} = 0 \Rightarrow \gamma_1 = 0$$

$$\text{Kurtosis } (\beta_2) = \frac{\mu_4}{\mu_2^2} = \frac{24}{2^2} = 6$$

$$\gamma_2 = \beta_2 - 3 = 6 - 3 = 3$$

Hence, it is concluded that from the values of γ_1 and γ_2 , the standard Laplace distribution is symmetrical and leptokurtic.

Moment Generating function of standard Laplace Distribution:

The mgf of double exponential function distribution is given by;

$$M_x(t) = E(e^{tx}) = \int_{-\infty}^{\infty} e^{tx} f(x) dx$$

$$= \int_{-\infty}^{\infty} e^{tx} \frac{1}{2} e^{-|x|} dx$$

$$= \frac{1}{2} \left[\int_{-\infty}^0 e^{tx} e^x dx + \int_0^{\infty} e^{tx} e^{-x} dx \right]$$

$$= \frac{1}{2} \left[\int_{-\infty}^0 e^{(1+t)x} dx + \int_0^{\infty} e^{-(1-t)x} dx \right]$$

$$= \frac{1}{2} \left[\left[\frac{e^{(1+t)x}}{1+t} \right]_{-\infty}^0 - \left[\frac{e^{-(1-t)x}}{1-t} \right]_0^{\infty} \right]$$

$$= \frac{1}{2} \left[\frac{1}{1+t} + \frac{1}{1-t} \right] = \frac{1}{2} \frac{1-t+1+t}{1-t^2}$$

$$= \frac{1}{1-t^2}$$

$$\therefore M_x(t) = (1-t^2)^{-1}$$

$$M_x(t) = 1 + t^2 + t^4 + \dots$$

$$= 1 + \frac{2! t^2}{2!} + \frac{4! t^4}{4!} + \dots$$

$$\mu_r' = \text{coefficient of } \frac{t^r}{r!} \text{ in } M_x(t)$$

$$\mu_1' = 0, \quad \mu_2' = 2! = 2$$

Raw moment

$$\mu_3' = 0, \quad \mu_4' = 4! = 4 \times 3 \times 2 \times 1 = 24$$

Central moment

$$\mu_1 = 0, \quad \mu_2 = 2, \quad \mu_3 = 0, \quad \mu_4 = 24$$

Characteristic function of Standard Laplace Distribution:

Distribution:

The characteristic function of standard Laplace distribution is given as,

$$\phi_x(t) = E(e^{itx}) = \int_{-\infty}^{\infty} e^{itx} f(x) dx$$

$$= \int_{-\infty}^{\infty} e^{itx} \frac{1}{2} e^{-|x|} dx$$

$$= \frac{1}{2} \left[\int_{-\infty}^0 e^{itx} e^{-x} dx + \int_0^{\infty} e^{itx} e^{-x} dx \right]$$

$$= \frac{1}{2} \left[\int_{-\infty}^0 e^{(1+it)x} dx + \int_0^{\infty} e^{-(1-it)x} dx \right]$$

$$= \frac{1}{2} \left[\left[\frac{e^{(1+it)x}}{(1+it)} \right]_{-\infty}^0 - \left[\frac{e^{-(1-it)x}}{(1-it)} \right]_0^{\infty} \right]$$

$$= \frac{1}{2} \left[\frac{1}{1+it} + \frac{1}{1-it} \right]$$

$$\boxed{\phi_x(t) = \frac{1}{1+t^2}}$$

$$\boxed{i^2 = -1}$$

$$\phi_x(t) = (1+t^2)^{-1}$$

$$\phi_x(t) = 1 - t^2 + t^4 - t^6 + \dots$$

$$\phi_x(t) = 1 + (it)^2 + (it)^4 + (it)^6 + \dots$$

$\therefore \mu_r' = \text{coefficient of } \frac{(it)^r}{r!} \text{ in } \phi_x(t)$

$$\therefore \mu_1' = 0, \mu_2' = 2, \mu_3' = 0, \mu_4' = 4! = 24$$

$$E(x) = 0, \quad V(x) = 2$$